Statistics & Data Analysis

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b, where $a > 0$ and $b > 0$.

Given the function $f(x) = a \cdot b^x$, and the points $(1, 12)$ and $(3, 108)$. From the point $(1, 12)$, we have: $12 = a \cdot b^1 \quad (1)$ From the point $(3, 108)$, we have: $108 = a \cdot b^3 \quad (2)$ Divide equation (2) by equation (1): $\frac{108}{12} = \frac{a \cdot b^3}{a \cdot b^1}$ $9 = b^2$ Since $b > 0$, we take the positive square root: $b = \sqrt{9} = 3$ Now substitute the value of b into equation (1): $12 = a \cdot 3^1$ $12 = 3a$ $a = \frac{12}{3} = 4$ Therefore, $a = 4$ and $b = 3$.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The first four terms of an arithmetic sequence are $7, 11, 15, 19, \ldots$. Which expression represents the nth term, a_n, of the sequence for all integers $n \geq 1$?

An arithmetic sequence has the general explicit formula $a_n = a_1 + (n - 1)d$, where a_1 is the first term and d is the common difference. From the given terms: 1. The first term is $a_1 = 7$. 2. The common difference is $d = 11 - 7 = 4$. Substituting these values into the formula: $a_n = 7 + (n - 1)(4) = 7 + 4n - 4 = 4n + 3$ Testing the first two terms: - For $n = 1$: $a_1 = 4(1) + 3 = 7$ - For $n = 2$: $a_2 = 4(2) + 3 = 11$ Therefore, the correct choice is $a_n = 4n + 3$. Common Misconceptions:

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

An arithmetic sequence is defined by its first term $a_1 = 7$ and its common difference d. The sum of the first n terms of this sequence is given by $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$. If the sum of the first 10 terms is $S_{10} = 115$, find the value of the common difference d.

We are given the first term $a_1 = 7$ and the sum of the first 10 terms $S_{10} = 115$. The formula for the sum of the first n terms of an arithmetic sequence is $S_n = \frac{n}{2}(2a_1 + (n - 1)d)$. Substitute the given values into the formula: $115 = \frac{10}{2}(2(7) + (10 - 1)d)$ $115 = 5(14 + 9d)$ Divide both sides by 5: $\frac{115}{5} = 14 + 9d$ $23 = 14 + 9d$ Subtract 14 from both sides: $23 - 14 = 9d$ $9 = 9d$ Divide by 9: $d = \frac{9}{9}$ $d = 1$ Therefore, the common difference d is 1.

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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The first three terms of an arithmetic sequence are given by $a_1 = 5$, $a_2 = 8$, and $a_3 = 11$. What is the 10th term of this sequence?

The given sequence is an arithmetic sequence with the first term $a_1 = 5$. The common difference d can be found by subtracting any term from its succeeding term: $d = a_2 - a_1 = 8 - 5 = 3$. The formula for the n-th term of an arithmetic sequence is $a_n = a_1 + (n - 1)d$. To find the 10th term ($n = 10$), we substitute the values: $a_{10} = 5 + (10 - 1) \times 3 = 5 + 9 \times 3 = 5 + 27 = 32$. Therefore, the 10th term is 32.

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the arithmetic sequence where the first term is 5 and the common difference is 3. What is the 4th term of this sequence?

The formula for the nth term of an arithmetic sequence is given by $a_n = a_1 + (n - 1)d$, where a_1 is the first term and d is the common difference. In this problem, we are given: $a_1 = 5$ $d = 3$ We need to find the 4th term, so $n = 4$. Substituting these values into the formula: $a_4 = 5 + (4 - 1) \times 3$ $a_4 = 5 + (3) \times 3$ $a_4 = 5 + 9$ $a_4 = 14$ Therefore, the 4th term of the sequence is 14.

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Simplify the expression: $\frac{2^5 \cdot 2^3}{2^4}$.

To simplify the expression $\frac{2^5 \cdot 2^3}{2^4}$, we use the properties of exponents. First, we combine the terms in the numerator by adding their exponents: $2^5 \cdot 2^3 = 2^{5+3} = 2^8$. Then, we divide this result by 2^4 by subtracting the exponents: $\frac{2^8}{2^4} = 2^{8-4} = 2^4$. Common errors might include adding exponents in the denominator or incorrectly applying the division rule.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

What is the solution to the equation $2^{x+1} = 32$?

To solve the exponential equation $2^{x+1} = 32$, express both sides with the same base: 1. Write 32 as a power of 2: $32 = 2^5$ 2. Substitute this into the equation: $2^{x+1} = 2^5$ 3. Since the bases are equal and positive (and not equal to 1), equate the exponents: $x + 1 = 5$ 4. Subtract 1 from both sides: $x = 4$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

A set of five exam scores has a mean of 84. Four of the scores are 78, 86, 92, and 80. Find the 5th score, and find the new mean if the lowest score is removed.

Total sum of the 5 scores = $5 \times 84 = 420$. Sum of the 4 known scores = $78 + 86 + 92 + 80 = 336$. 5th score = $420 - 336 = 84$. Removing the lowest score (78) leaves four scores: 80, 84, 86, 92. New sum = $420 - 78 = 342$. New mean = $342/4 = 85.5$ (an increase of 1.5 points).

9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A distribution of test scores has first quartile $Q_1 = 58$ and third quartile $Q_3 = 82$. Using the standard $1.5 \times \text{IQR}$ rule, what is the lowest score that would NOT be considered an outlier?

First compute the interquartile range (IQR): $\text{IQR} = Q_3 - Q_1 = 82 - 58 = 24$. Calculate the lower fence below which values are classified as low outliers: $\text{Lower Fence} = Q_1 - 1.5(\text{IQR}) = 58 - 1.5(24) = 58 - 36 = 22$. Any score at or above 22 is not a low outlier. The lowest non-outlier score is 22.

10 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)**

[Statistics & Data Analysis · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Level 3 · AdvancedIn the hard Module 2, questions 18–22 are typically high-difficulty multi-step problems with parameters k. If a student has 7 minutes remaining for the last 3 questions, which approach optimizes score yield: attempting all 3 partially, or using Desmos regression and choice elimination to guarantee complete accuracy on 2 of them?

Optimal Strategy: On the Digital SAT's Item Response Theory (IRT) scoring model, fully correct answers yield substantially higher raw theta weights than rushed guesses. Allocating 3.5 minutes each to 2 questions using Desmos graphing/regression guarantees 2 correct hard items (+40–60 scaled points), while placing a rapid educated guess on the 3rd question preserves time with zero guessing penalty.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

For what value of k will the system have no solution?

$\{3x - 2y = 8;\ 6x + ky = 5$ Solution: 1. A linear system has no solution if the lines are parallel (identical slopes, different y-intercepts). 2. Express the first line in slope-intercept form: $-2y = -3x + 8 \Rightarrow y = \frac{3}{2}x - 4 \quad \Rightarrow m_1 = \frac{3}{2}$. 3. Express the second line in slope-intercept form: $ky = -6x + 5 \Rightarrow y = -\frac{6}{k}x + \frac{5}{k} \quad \Rightarrow m_2 = -\frac{6}{k}$. 4. Equate slopes to make lines parallel.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the equation $\sqrt{2x+7} = x+2$ for all real values of x.

Square both sides of the equation: $(\sqrt{2x+7})^2 = (x+2)^2 \Rightarrow 2x+7 = x^2+4x+4$. Rearrange into standard quadratic form: $x^2+2x-3=0 \Rightarrow (x+3)(x-1)=0 \Rightarrow x=1$ or $x=-3$. Now check both candidate roots in the original equation:For $x=1$: LHS = $\sqrt{2(1)+7} = \sqrt{9} = 3$; RHS = $1+2=3$. (Valid ✓)For $x=-3$: LHS = $\sqrt{2(-3)+7} = \sqrt{1} = 1$; RHS = $-3+2=-1$. Since $1eq-1$, $x=-3$ is extraneous.The only real solution is.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Evaluate the numerical value of $16^{-3/4} + \left(\frac{27}{8}\right)^{-2/3}$.

1. Evaluate the first term: $16^{-3/4} = \frac{1}{16^{3/4}} = \frac{1}{(\sqrt[4]{16})^3} = \frac{1}{2^3} = \frac{1}{8}$. 2. Evaluate the second term: inverting the fraction removes the negative exponent: $\left(\frac{27}{8}\right)^{-2/3} = \left(\frac{8}{27}\right)^{2/3} = \frac{(\sqrt[3]{8})^2}{(\sqrt[3]{27})^2} = \frac{2^2}{3^2} = \frac{4}{9}$. 3. Combine with a common denominator of 72: $\frac{1}{8} + \frac{4}{9} = \frac{9}{72} + \frac{32}{72} = \frac{41}{72}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.01 Systems & Inequalities

Find the sum of all integers between 10 and 100 (inclusive) that are divisible by 4.

1. Identify the sequence: The first multiple of $4 \geq 10$ is $a_1 = 12$. The last multiple ≤ 100 is $a_n = 100$. The common difference is $d = 4$. 2. Find the number of terms n: $a_n = a_1 + (n-1)d \Rightarrow 100 = 12 + (n-1)4 \Rightarrow 88 = 4(n-1) \Rightarrow n-1 = 22 \Rightarrow n = 23$. 3. Calculate the sum using $S_n = \frac{n}{2}(a_1 + a_n)$: $S_{23} = \frac{23}{2}(12+100) = \frac{23}{2}(112) = 23 \cdot 56 = 1288$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 7.01 Sequences & Series

Find the equation of the line that passes through $(4, -1)$ and is perpendicular to the line $2x - 3y = 9$. Express your answer in slope-intercept form.

1. Find the slope of the given line by converting to slope-intercept form: $-3y = -2x + 9 \Rightarrow y = \frac{2}{3}x - 3$. The slope of the given line is $m_1 = \frac{2}{3}$. 2. The perpendicular slope is the negative reciprocal: $m_\perp = -\frac{3}{2}$. 3. Use point-slope form with $(x_1, y_1) = (4, -1)$: $y - (-1) = -\frac{3}{2}(x-4) \Rightarrow y + 1 = -\frac{3}{2}x + 6 \Rightarrow y = -\frac{3}{2}x + 5$.

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17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 8.01 Factoring Polynomials**

A projectile is launched from ground level with height modelled by $h(t) = -16t^2 + 64t + 80$ feet after t seconds. Find (a) the maximum height reached, and (b) the time when it hits the ground.

1. The maximum occurs at the vertex time $t = -\frac{b}{2a}$: $t = -\frac{64}{2(-16)} = \frac{64}{32} = 2$ seconds. Substitute $t = 2$ to get maximum height: $h(2) = -16(2^2) + 64(2) + 80 = -64 + 128 + 80 = 144$ feet. 2. Ground impact occurs when $h(t) = 0$: $-16t^2 + 64t + 80 = 0 \Rightarrow t^2 - 4t - 5 = 0 \Rightarrow (t - 5)(t + 1) = 0$. Since time must be non-negative, $t = 5$ seconds.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 9.01 Radicals & Exponents**

In a geometric sequence, the 3rd term is 18 and the 6th term is 486. Find the common ratio r, the first term a_1, and the sum of the first 5 terms S_5.

Use the general formula $a_n = a_1 r^{n-1}$: $a_3 = a_1 r^2 = 18, \quad a_6 = a_1 r^5 = 486$. Divide a_6 by a_3: $\frac{a_1 r^5}{a_1 r^2} = r^3 = \frac{486}{18} = 27 \Rightarrow r = 3$. Solve for a_1: $a_1(3^2) = 18 \Rightarrow 9a_1 = 18 \Rightarrow a_1 = 2$. Compute S_5 using the geometric series sum formula: $S_5 = \frac{a_1(r^5 - 1)}{r - 1} = \frac{2(3^5 - 1)}{3 - 1} = \frac{2(243 - 1)}{2} = 242$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 1.01 Linear Equations & Lines**

Solve the nonlinear system $y = x^2 - 3x + 2$ and $y = 2x - 4$ algebraically. State all intersection points (x, y).

Equate the expressions for y: $x^2 - 3x + 2 = 2x - 4 \Rightarrow x^2 - 5x + 6 = 0$. Factor the quadratic: $(x - 2)(x - 3) = 0 \Rightarrow x = 2$ or $x = 3$. Find corresponding y-values using $y = 2x - 4$:For $x = 2$: $y = 2(2) - 4 = 0$, yielding intersection point $(2, 0)$.For $x = 3$: $y = 2(3) - 4 = 2$, yielding intersection point $(3, 2)$.The system intersects at $(2, 0)$ and $(3, 2)$.

20 **[Challenging]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)** • **Cross-Lesson: 10.01 Rational Expressions**

Find the inverse function $f^{-1}(x)$ for $f(x) = \frac{3x + 4}{2x - 5}$, and state the domain of f^{-1}.

Let $y = \frac{3x + 4}{2x - 5}$. Swap variables: $x = \frac{3y + 4}{2y - 5}$. Multiply both sides by $2y - 5$: $x(2y - 5) = 3y + 4 \Rightarrow 2xy - 5x = 3y + 4$. Group all y-terms on one side: $2xy - 3y = 5x + 4 \Rightarrow y(2x - 3) = 5x + 4$. Divide to isolate y: $f^{-1}(x) = \frac{5x + 4}{2x - 3}$. Domain of f^{-1}: $x eq \frac{3}{2}$ (all real numbers except $x = 1.5$).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for review: operations with rational numbers consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for units resulting from a formula or graph consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

23 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The bar chart below shows the frequency distribution of an initial dataset of 10 test scores. Two additional scores, a and b, are added to this dataset to form a new dataset of 12 scores. The mean of the new dataset is 10, and its population variance is 26. What is the absolute difference $|a - b|$ between the two added scores? *(Note: The population variance of N values $y_1, y_2, \ldots, y_N$ with mean μ is $\sigma^2 = \frac{1}{N}\sum_{i=1}^{N}(y_i - \mu)^2$.)*

a. 8 ✓

b. 6

c. 10

d. 12

1. **Analyze the initial dataset of $n = 10$ scores from the bar chart:** - The scores and their frequencies are: 4 (freq 1), 6 (freq 1), 8 (freq 6), 10 (freq 1), and 12 (freq 1). - By symmetry about 8, the mean of the initial dataset is: $\mu_1 = \frac{1(4) + 1(6) + 6(8) + 1(10) + 1(12)}{10} = \frac{80}{10} = 8$ - The sum of the squares of the initial observations is: $\sum_{i=1}^{10} x_i^2 = 1(4^2) + 1(6^2) + 6(8^2) + 1(10^2) + 1(12^2) = 16 + 36 + 384 + 100 + 144 = 680$ 2. **Set up the equations for the.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A small company manufactures two types of artisanal cheeses: 'Mild Cheddar' and 'Sharp Provolone'. The production process involves two main stages: milk preparation and aging. The time (in hours) required for each kilogram of cheese at each stage is given in the table below. | Cheese Type | Milk Prep (hours/kg) | Aging (hours/kg) | |----------------|----------------------|------------------| | Mild Cheddar | 0.5 | 1.0 | | Sharp Provolone | 1.2 | 2.5 | Due to resource constraints, the company has a maximum of 120 hours available for milk preparation and 280 hours for aging per week. The profit per kilogram of Mild Cheddar is \$15. Let x be the number of kilograms of Mild Cheddar produced per week, and y be the number of kilograms of Sharp Provolone produced per week. Formulate a system of linear inequalities representing the production constraints and the objective function to maximize the total weekly profit. Which of the following options correctly represents the feasible region and the objective function?

a. **Constraints:** $0.5x + 1.2y \leq 120$, $1.0x + 2.5y \leq 280$, $x \geq 0$, $y \geq 0$. **Objective function:** $P = 8x + 15y$. ✓

b. Constraints: $0.5x + 1.0y \leq 120$, $1.2x + 2.5y \leq 280$, $x \geq 0$, $y \geq 0$. Objective function: $P = 15x + 8y$.

c. Constraints: $0.5x + 1.2y \geq 120$, $1.0x + 2.5y \geq 280$, $x \geq 0$, $y \geq 0$. Objective function: $P = 8x + 15y$.

d. Constraints: $0.5x + 1.0y \leq 120$, $1.2x + 2.5y \leq 280$, $x \geq 0$, $y \geq 0$. Objective function: $P = 8x + 15y$.

The problem requires formulating a system of linear inequalities and an objective function based on given constraints and profit margins. Let x represent the kilograms of Mild Cheddar and y represent the kilograms of Sharp Provolone. Milk Preparation Constraint: The total time spent on milk preparation cannot exceed 120 hours. For Mild Cheddar, it takes 0.5 hours/kg, and for Sharp Provolone, it takes 1.2 hours/kg. So, the inequality is $0.5x + 1.2y \leq 120$. Aging Constraint: The total time spent on aging cann.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A student has scores of 81, 85, and 88 on three mathematics tests. What score must the student earn on the fourth test to have an average (arithmetic mean) score of exactly 87 across all four tests?

a. 87

b. 91

c. 94 ✓

d. 98

To find the required score on the fourth test, let x represent that score. The arithmetic mean of the four test scores is given by: $\frac{81 + 85 + 88 + x}{4} = 87$ First, multiply both sides of the equation by 4 to determine the total points required across all four tests: $81 + 85 + 88 + x = 87 \times 4$ $254 + x = 348$ Next, subtract 254 from both sides to solve for x: $x = 348 - 254 = 94$ Thus, the student must earn a score of 94 on the fourth test. Common Misconceptions: - Choice A (87) incorrectl.

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26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The table below shows the results of a survey on the preferred music genres of students in two different schools, School A and School B. The total number of students surveyed in School A is 200, and the total number of students surveyed in School B is 150. | Genre | School A | School B | |------------|----------|----------| | Pop | 70 | 55 | | Rock | 60 | 45 | | Electronic | 40 | 30 | | Classical | 30 | 20 | If a student is chosen at random from School A, what is the probability that they prefer Electronic music?

a. **0.20** ✓

b. 0.33

c. 0.25

d. 0.15

The question asks for the probability that a randomly chosen student from School A prefers Electronic music. First, identify the number of students in School A who prefer Electronic music from the table. This number is 40. Next, identify the total number of students surveyed in School A. This number is given as 200. To find the probability, divide the number of students who prefer Electronic music by the total number of students in School A: P(Electronic | School A) = (Number of students in School A who prefer Elec.

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A box plot shows Q1=20, Median=30, Q3=45. What is the IQR?

a. 15

b. 25 ✓

c. 10

d. 45

Applying Statistics & Data Analysis principles, IQR = Q3 – Q1 = $45 - 20 = 25$. Therefore, the verified correct option is 25.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the median of the data set: $4, 6, 9, 20, 36, 40$.

a. 9

b. $\frac{31}{2}$

c. $\frac{29}{2}$ ✓

d. $\frac{27}{2}$

With 6 ordered values, the median is the average of the 3rd and 4th: $\frac{9+20}{2} = \frac{29}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set has values $5, 7, 7, 8, 10$. What is the mean absolute deviation?

a. 1.5

b. 0.8

c. 1.2 ✓

d. 2.0

Mean=7.4; deviations: 2.4,0.4,0.4,0.6,2.6; MAD=$\frac{6.4}{5} \approx 1.28 \approx 1.2$ (closest).

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The standard deviation measures:

a. The most common value

b. Center of data

c. **Spread of data around the mean** ✓

d. The range

Standard deviation measures how spread out values are from the mean.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the mean of the data set: $3, 8, 9, 12, 34$.

a. $\frac{71}{5}$

b. 9

c. $\frac{66}{5}$ ✓

d. $\frac{137}{10}$

Mean $= \frac{3+8+9+12+34}{5} = \frac{66}{5} = \frac{66}{5}$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A sample of 10 students scored: 70,75,80,80,85,85,85,90,90,100. What is the median?

a. 80

b. 84

c. 85 ✓

d. 87.5

Ordered list, 5th and 6th values: 85 and 85. Median = 85.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A survey asks 50 students their favorite subject. This data is:

a. Ratio

b. Quantitative

c. Continuous

d. **Categorical** ✓

Subjects are categories, not numbers → categorical data.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Which measure of center is most affected by outliers?

a. Median

b. **Mean** ✓

c. Mode

d. Range

The mean is sensitive to extreme values (outliers).

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A student scores 10 (1 times), 13 (1 times), 24 (1 times), 29 (3 times). What is the weighted average score?

a. $\frac{67}{3}$ ✓

b. $\frac{137}{6}$

c. $\frac{70}{3}$

d. $\frac{64}{3}$

Weighted mean $= \frac{134}{6} = \frac{67}{3}$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which type of correlation does a scatter plot show if both variables increase together?

a. No correlation

b. **Positive** ✓

c. Negative

d. Inverse

Both increase together → positive correlation.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In a normal distribution, approximately what percent of data falls within 1 standard deviation of the mean?

a. 50%

b. 99%

c. 95%

d. **68%** ✓

The 68-95-99.7 rule: 68% within $\pm 1\sigma$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 9.01 Radicals & Exponents

Which measure is not affected by outliers?

a. Mean

b. Range

c. **Median ✓**

d. Standard deviation

The median is resistant to extreme values.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Linear Equations & Lines

A researcher is studying the relationship between the number of hours spent studying for a statistics exam and the score achieved. A random sample of 10 students yielded the following data, where x is the number of hours studied and y is the exam score: | Hours Studied (x) | Exam Score (y) | |---|---| | 2 | 65 | | 3 | 70 | | 4 | 78 | | 5 | 82 | | 6 | 88 | | 7 | 91 | | 8 | 95 | | 9 | 98 | | 10 | 100 | | 11 | 103 | The researcher models the relationship using a linear regression line of the form $y = \hat{a}x + \hat{b}$. Calculate the predicted exam score for a student who studies for $x = 7.5$ hours, given that the calculated regression line is $y = 4.5x + 55$.

a. **97.5 ✓**

b. 95.0

c. 100.0

d. 92.5

The problem provides a linear regression model $\hat{y} = 5x + 60$, where x is the number of hours studied and $\hat{y}$ is the predicted exam score. We are asked to find the predicted exam score for a student who studies for $x = 7.5$ hours. To find the predicted exam score, we substitute $x = 7.5$ into the regression equation: $\hat{y} = 5(7.5) + 60$ First, calculate the product of 5 and 7.5: $5 \times 7.5 = 37.5$ Now, add 60 to this result: $\hat{y} = 37.5 + 60$ $\hat{y} = 97.5$ Therefore, the predicted exam s.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 10.01 Rational Expressions

Consider the following set of 11 exam scores listed in ascending order:

$37, 38, 40, 43, 45, 47, 47, 51, 51, 55, 56$ What is the interquartile range (IQR) of this data set?

a. 11 ✓

b. 14

c. 47

d. 19

For 11 ordered data points, the median (Q_2) is the 6th value: 47. The lower half of the data consists of the first 5 values: $[37, 38, 40, 43, 45]$, with lower quartile $Q_1 = 40$ (the median of the lower half). The upper half consists of the last 5 values: $[47, 51, 51, 55, 56]$, with upper quartile $Q_3 = 51$ (the median of the upper half). The interquartile range (IQR) is: $\text{IQR} = Q_3 - Q_1 = 51 - 40 = 11$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 2.01 Quadratic Functions

The table shows 5 students' exam scores. What is the average (arithmetic mean) of the data?

a. 82

b. 84

c. **86** ✓

d. 88

$\frac{82 + 91 + 76 + 88 + 93}{5} = \frac{430}{5} = 86.$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.01 Exponents & Polynomials

Let $x, y,$ and z be real numbers satisfying the system of equations $\{x + y + z = 3; x^2 + y^2 + z^2 = 9$ What is the minimum possible value of $x^3 + y^3 + z^3$?

a. 15 ✓

b. 21

c. 27

d. 33

We use elementary symmetric polynomials to express $x^3 + y^3 + z^3$ in terms of xyz. First, square the first equation: $(x + y + z)^2 = x^2 + y^2 + z^2 + 2(xy + yz + zx)$ Substituting the given values:

$3^2 = 9 + 2(xy + yz + zx) \Rightarrow 2(xy + yz + zx) = 0 \Rightarrow xy + yz + zx = 0$ Next, recall the algebraic identity:

$x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - (xy + yz + zx))$ Substituting our known values:

$x^3 + y^3 + z^3 - 3xyz = 3(9 - 0) = 27 \Rightarrow x^3 + y^3 + z^3 = 27 + 3xyz$ To m.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Functions & Transformations

What is the complete solution set of the inequality $\log_{2x}(x^2 - 5x + 6) \leq 1$?

a. $(0, \frac{1}{2}) \cup [1, 2) \cup (3, 6]$ ✓

b. $[1, 2) \cup (3, 6]$

c. $(0, \frac{1}{2}) \cup [1, 6]$

d. $(0, \frac{1}{2}) \cup [1, 2] \cup [3, 6]$

To solve the inequality $\log_{2x}(x^2 - 5x + 6) \leq 1$, we must analyze the domain restrictions and split into cases based on the variable base $2x$. **Step 1: Determine the Domain** 1. **Base conditions:** The base must be strictly positive and not equal to 1: $2x > 0 \Rightarrow x > 0 \quad \text{and} \quad 2xe1 \Rightarrow xe\frac{1}{2}$. 2. **Argument condition:** The argument must be strictly positive: $x^2 - 5x + 6 > 0 \Leftrightarrow (x - 2)(x - 3) > 0 \Leftrightarrow x3$. Combining these conditions.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

In the figure, ΔABC is a right triangle with the right angle at vertex C, and altitude CD is drawn perpendicular to the hypotenuse AB. The lengths of the segments are given in terms of a variable x by $AD = x + 2$, $DB = 3x - 5$, and $CD = x + 5$. What is the perimeter of ΔABC?

a. 35

b. 50

c. 60 ✓

d. 150

We solve the problem through the following steps: 1. **Apply the Geometric Mean Theorem (Right-Triangle Altitude Theorem):** In a right triangle with an altitude drawn to the hypotenuse, the length of the altitude is the geometric mean of the two segments of the hypotenuse: $CD^2 = AD \cdot DB$ 2. **Formulate the quadratic equation:** Substitute the given algebraic expressions into the theorem: $(x + 5)^2 = (x + 2)(3x - 5)$ 3. **Expand and simplify to standard quadratic form:** $x^2 + 10x + 25 = 3x^2 + x - 10$ Rear.

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45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Systems & Inequalities

Consider the system of inequalities: $\{x^2 + y^2 \leq 16; |x| + |y| \geq 4$ Which of the following statements best describes the region defined by this system? A. The region inside a circle of radius 4, excluding the interior of a square with vertices at $(\pm 4, 0)$ and $(0, \pm 4)$. B. The region inside a circle of radius 4, including the boundary of a square with vertices at $(\pm 4, 0)$ and $(0, \pm 4)$. C. The region outside a circle of radius 4, excluding the interior of a square with vertices at $(\pm 4, 0)$ and $(0, \pm 4)$. D. The region between a circle of radius 4 and a square with vertices at $(\pm 4, 0)$ and $(0, \pm 4)$.

a. **A ✓**

b. B

c. C

d. D

The first inequality, $x^2 + y^2 \leq 16$, represents the region inside and on the circle centered at the origin with a radius of 4. The second inequality, $|x| + |y| \geq 4$, represents the region outside and on the boundary of a square whose vertices are $(\pm 4, 0)$ and $(0, \pm 4)$. This square is formed by the lines $x + y = 4$, $x - y = 4$, $-x + y = 4$, and $-x - y = 4$ in the respective quadrants. We are looking for the intersection of these two regions. This means we want the points that are both inside or on the circle o.

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